

Post-Quantum Cryptography: A Comprehensive Survey

MA2209 - Number Theory and Cryptography

Group Project Report

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Abstract

This comprehensive report integrates the collective research of our group project on Post-Quantum Cryptography (PQC). We trace the evolution of cryptographic security from classical public-key systems, examining how Shor's quantum algorithm invalidates their foundational hardness assumptions. We introduce the landscape of post-quantum alternatives, explore the key number-theoretic building blocks of the five main PQC families, and review the finalized NIST standardization outcomes as of 2026. Furthermore, we detail the rigorous analytical methods used to evaluate PQC schemes, catalog the present shortcomings and implementation hazards of the transition, and outline the future prospects, mathematical advances, and technical optimizations that will define the next decade of quantum-resistant security.

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1 Introduction and Classical Cryptography

Modern secure communication rests on asymmetric (public-key) cryptography, first formalized by Diffie and Hellman in 1976. Two mathematically related keys are generated—a public key shared openly, and a private key kept secret—such that encrypting with one key can only be reversed with the other.

1.1 Hardness Assumptions Underlying Classical Security

Three mathematical problems have historically underpinned public-key cryptography:

1. **Integer Factorisation Problem (IFP):** Given a large composite $N = p \cdot q$ with p, q large primes, recover p and q . This is the foundation of RSA.
2. **Discrete Logarithm Problem (DLP):** Given g, p , and $y = g^x \pmod{p}$, recover x . This underpins classical Diffie-Hellman and the Digital Signature Algorithm (DSA).
3. **Elliptic Curve Discrete Logarithm Problem (ECDLP):** Given points P and $Q = kP$ on an elliptic curve, recover the scalar k . This is the basis of ECDH and ECDSA.

1.2 RSA: The Cornerstone of Classical Public-Key Cryptography

RSA relies on Euler's Totient Function, where for $n = p \cdot q$, $\phi(n) = (p-1)(q-1)$. Any integer a with $\gcd(a, n) = 1$ satisfies $a^{\phi(n)} \equiv 1 \pmod{n}$.

- **Encryption:** $c = m^e \pmod{n}$
- **Decryption:** $m = c^d \pmod{n}$
- **Signatures:** $\sigma = \mathcal{H}(m)^d \pmod{n}$

An adversary who can factor n immediately recovers $\phi(n)$ and hence the private key d . The best known classical algorithm is the General Number Field Sieve (GNFS), with sub-exponential complexity:

$$L_n \left[\frac{1}{3}, \sqrt[3]{\frac{64}{9}} \right] = \exp \left(\left(\sqrt[3]{\frac{64}{9}} + o(1) \right) (\ln n)^{1/3} (\ln \ln n)^{2/3} \right)$$

For a 2048-bit modulus, this remains computationally infeasible classically.

1.3 Shor's Algorithm and the Collapse of Classical Hardness

Shor's algorithm (1994) reduces integer factorization to a period-finding problem: find the order r of $a \pmod{n}$ such that $a^r \equiv 1 \pmod{n}$. By preparing a quantum superposition and applying the Quantum Fourier Transform (QFT), a quantum computer extracts this period efficiently.

Shor's algorithm runs in polynomial time, collapsing the security of RSA, DH, DSA, ECDH, and ECDSA. Conversely, symmetric ciphers (AES) and hashes (SHA) are only weakened by Grover's algorithm (quadratic speedup) and survive by doubling parameter sizes.

Algorithm	Type	Complexity
Trial Division	Classical	$O(\sqrt{n})$
GNFS (best classical)	Classical	$L_n[1/3, c]$ (sub-exponential)
Shor’s Algorithm	Quantum	$O((\log n)^2 \log \log n \cdot \log \log \log n)$

Table 1: Complexity comparison for Integer Factorisation.

1.4 The “Harvest Now, Decrypt Later” Threat

Adversaries are actively collecting encrypted network traffic today with the intent to decrypt it once cryptographically relevant quantum computers (CRQCs) become available. Data requiring long-term confidentiality (government, medical, financial) is already at risk.

2 Key Number-Theoretic Building Blocks and NIST Standards

Post-Quantum Cryptography (PQC) shifts the hardness foundation to alternative problems for which no efficient quantum algorithm exists, while maintaining practical efficiency on classical hardware.

2.1 Major PQC families

2.1.1 Lattice-Based Cryptography

Lattice-based cryptography is the most prominent and practically deployed PQC family. It dominates the NIST standardisation outcome, providing both key encapsulation and digital signature schemes. A lattice is a discrete, periodic set of points in n -dimensional Euclidean space.

A lattice $\Lambda \subset \mathbb{R}^n$ is defined as $\Lambda(B) = \{z_1 b_1 + \dots + z_n b_n \mid z_i \in \mathbb{Z}\}$ for a basis matrix B .

- **Shortest Vector Problem (SVP):** Find the shortest non-zero lattice vector $\lambda_1(\Lambda) = \min_{v \in \Lambda \setminus \{0\}} \|v\|$.
- **Learning With Errors (LWE):** Given $A \in \mathbb{Z}_q^{m \times n}$ and $b = As + e \pmod{q}$ (with small noise e), recover s . Regev (2005) proved solving LWE on average is at least as hard as solving worst-case lattice problems.

NIST Standards: FIPS 203 (ML-KEM / Kyber), FIPS 204 (ML-DSA / Dilithium), FIPS 206 (FN-DSA / Falcon)

2.1.2 Code-Based Cryptography

Code-based cryptography derives its security from the theory of error-correcting codes — a field originally developed for reliable data transmission over noisy channels. It has the longest security history of any PQC family: the McEliece cryptosystem was proposed in 1978 and has never been broken.

Security derives from the NP-hard Syndrome Decoding Problem: Given a parity-check matrix H and a syndrome $s = Hx \pmod{2}$, find a binary vector x of small Hamming weight $\leq t$. The McEliece cryptosystem masks a structured code (e.g., Goppa code) using a random-looking disguise ($G' = SGP$). It boasts a 40-year security record but requires massive public keys.

NIST Status: HQC (Hamming Quasi-Cyclic) was selected in 2025 as a KEM backup.

2.1.3 Hash-Based Cryptography

Hash-based schemes are the most conservative PQC choice. Unlike other families, they require no algebraic structure — security reduces entirely to the assumed properties of a cryptographic hash function. If SHA-256 is secure, the signature scheme is secure.

It requires minimal assumptions: only preimage and collision resistance of hash functions. They utilize Merkle trees to aggregate One-Time Signatures (OTS). Grover’s algorithm is mitigated by using ≥ 256 -bit hash outputs.

NIST Standards: FIPS 205 (SLH-DSA / SPHINCS+), a stateless backup signature scheme.

2.1.4 Multivariate-Quadratic (MQ) Cryptography

Multivariate cryptography is based on the hardness of solving systems of nonlinear polynomial equations over finite fields — a class of problems with deep roots in algebraic geometry and computational complexity theory.

Its security is based on the NP-hard MQ problem: finding a vector $(x_1, \dots, x_n)$ satisfying m quadratic polynomials $P_i(x_1, \dots, x_n) = 0$ over a finite field $\mathbb{F}_q$. While offering tiny signatures, parameter sensitivity is high (e.g., the Rainbow break in 2022 via rank attacks). No MQ scheme was standardized in NIST Round 3.

2.1.5 Isogeny-Based Cryptography

Isogeny-based cryptography is the newest PQC family, building on the rich theory of elliptic curves. Rather than using points on a single elliptic curve (as in classical ECC), it exploits maps between different elliptic curves.

It exploits maps (isogenies) $\phi : E_1 \rightarrow E_2$ between supersingular elliptic curves defined over $\mathbb{F}_{p^2}$. The endomorphism rings are non-commutative quaternion algebras.

Status: The prominent candidate SIKE was broken in 2022 exploiting auxiliary torsion-point information. SQIsign remains an active research area offering the most compact keys independent of lattices.

2.2 NIST Standardization (as of 2026)

NIST’s Post-Quantum Cryptography Standardization process has produced the following core standards (published in 2024):

- **FIPS 203:** ML-KEM (Module-Lattice-based Key Encapsulation Mechanism, derived from CRYSTALS-Kyber) — primary standard for key establishment.
- **FIPS 204:** ML-DSA (Module-Lattice-based Digital Signature Algorithm, derived from CRYSTALS-Dilithium) — primary standard for digital signatures.
- **FIPS 205:** SLH-DSA (Stateless Hash-based Digital Signature Algorithm, derived from SPHINCS+) — backup signature scheme for mathematical diversity.

Additional standards in development:

- **FIPS 206** (FN-DSA/Falcon): Compact lattice-based signatures.
- **HQC:** Code-based KEM selected in March 2025; draft expected around 2026–2027.

NIST encourages hybrid schemes (classical + PQC) during the transition period and emphasizes crypto-agility.

Family	Hard Problem	Strength	Weakness	Examples
Lattice	SVP, LWE, MLWE	Efficient, versatile; NIST standard	Moderate key/CT size	ML-KEM, ML-DSA
Code	Syndrome Decoding	40+ year history; NP-hard	Very large public keys	Classic McEliece, HQC
Hash	Preimage/Collision	Minimal assumptions	Large signatures	SLH-DSA
Multivariate	MQ-polynomials	Very small, fast sigs	Structural breaks	Rainbow (broken)
Isogeny	CIP, SSDDH	Smallest key sizes	SIKE broken (2022)	SQIsign

Table 2: Mathematical Family Comparison.

3 Analysing PQC: Methods and Metrics

3.1 Security Reductions & Parameter Selection

A security reduction proves that breaking a scheme is as hard as solving a well-studied problem. ML-KEM relies on a tight reduction from worst-case SVP to average-case MLWE in the Quantum Random Oracle Model (QROM).

Parameters are matched to NIST security levels:

- **Level 1:** Equivalent to AES-128 exhaustive search.
- **Level 3:** Equivalent to AES-192 exhaustive search.
- **Level 5:** Equivalent to AES-256 exhaustive search.

For lattice schemes, the Core-SVP Model sets security based on the cost of the Block Korkine-Zolotarev (BKZ) algorithm. A quantum BKZ attack costs $2^{0.265 \cdot \beta}$. ML-KEM-768 utilizes $n = 256, q = 3329, k = 3$, requiring a BKZ block size $\beta = 177$, yielding ~ 164 -bit quantum security (comfortably NIST Level 3).

3.2 Performance Metrics

The shift to PQC introduces significant size overheads. Table 3 illustrates size variations, and Table 4 details performance speeds.

3.3 Attack Models and Mathematical Tools

PQC schemes are analyzed against multiple threat vectors:

- **Quantum Attacks:** Grover’s algorithm halves bit-security for hashes but only provides a $\sim 10\%$ bit reduction against BKZ lattice sieving (handled by slightly larger parameters). No Shor-like exponential speedup exists for SVP, LWE, or SDP.
- **Structural Breaks:** Auxiliary torsion-point data published in the SIKE protocol enabled a classical break via the Kani lemma. Rainbow fell to a rank attack exploiting its oil-and-vinegar structure over finite fields.

Scheme	Family	Pub Key	Priv Key	Sig/CT	Level
RSA-2048	IFP	256 B	~1,200 B	256 B	-
ECDSA P-256	ECDLP	64 B	32 B	64 B	-
ML-KEM-768	Lattice	1,184 B	2,400 B	1,088 B	3
ML-DSA-65	Lattice	1,952 B	4,000 B	3,293 B	3
FN-DSA-512	Lattice	897 B	1,281 B	666 B	1
SLH-DSA-128f	Hash	32 B	64 B	17,088 B	1
Classic McEliece	Code	261,120 B	6,492 B	128 B	1
SQIsign	Isogeny	~64 B	~782 B	~177 B	1

Table 3: Size comparison. B = bytes; CT = ciphertext.

Scheme	KeyGen (kc)	Enc/Sign (kc)	Dec/Verify (kc)
ML-KEM-768	~47	~56	~59
ML-DSA-65	~125	~195	~122
FN-DSA-512	~8,000	~900	~43
SLH-DSA-128f	~900	~10,000	~420
Classic McEliece	~1,500,000	~120	~130

Table 4: Approximate speed in thousands of cycles (kc) on x86-64 + AVX2.

- **Side-Channel Attacks:** PQC requires strict constant-time implementations. NTT butterfly operations on embedded devices are vulnerable to power analysis (SPA/DPA), and table lookups are vulnerable to cache attacks.

Mathematical analysis relies heavily on the **Number Theoretic Transform (NTT)** for $O(n \log n)$ polynomial multiplication, the **Gaussian Heuristic** ($\lambda_1(\Lambda) \approx \sqrt{n/(2\pi e)} \cdot \text{vol}(\Lambda)^{1/n}$), and **Gröbner Bases** for MQ analysis.

4 Present Shortcomings of Post-Quantum Cryptography

While post-quantum cryptography (PQC) represents a critical leap forward in securing digital communications against quantum threats, the field is far from mature. The transition from classical cryptographic systems such as RSA and ECC to PQC schemes introduces a range of technical, operational, and theoretical challenges. This section examines the principal shortcomings of current PQC approaches, drawing attention to size and performance overhead, new attack surfaces, implementation hazards, ecosystem diversity risks, migration complexity, and unresolved theoretical gaps.

4.1 Size and Performance Overhead

One of the most significant practical drawbacks of PQC algorithms is their large key sizes, ciphertext sizes, and computational overhead when compared to traditional public-key cryptographic systems.

The problem becomes more severe in code-based cryptography. Schemes such as Classic McEliece require public keys that may reach hundreds of kilobytes or even megabytes in size. While such systems possess strong historical confidence and resistance against known

attacks, their large memory requirements make them difficult to deploy in bandwidth-constrained environments.

This overhead introduces serious challenges for:

- Internet of Things (IoT) devices
- Smart cards and embedded systems
- Low-power sensors
- Mobile devices
- Real-time communication systems

Many of these systems operate under strict memory, storage, and energy constraints. The polynomial arithmetic used in lattice-based schemes, particularly operations involving Number Theoretic Transforms (NTTs), increases computational complexity and energy consumption. Thus, although PQC schemes provide quantum resistance, their performance costs remain a major barrier to widespread deployment, especially in constrained computing environments.

4.2 New and Expanded Attack Surfaces

Post-quantum cryptographic systems introduce new mathematical structures that, while enabling efficiency, also create entirely new categories of attack surfaces.

Many PQC algorithms are based on structured mathematical problems involving polynomial rings, modules, lattices, finite fields, or isogeny graphs. The additional algebraic structure present in these systems can sometimes be exploited by adversaries in ways that are not possible in more general mathematical settings.

Multivariate cryptographic systems also face long-standing risks from algebraic attacks. These attacks attempt to solve systems of multivariate polynomial equations efficiently, potentially compromising poorly chosen parameter sets.

Another major concern involves the future evolution of quantum algorithms themselves. Current security estimates are based on known classical and quantum attack techniques. However, future advances in quantum computing or cryptanalysis may significantly reduce the security margins of existing PQC systems. As a result, the long-term stability of PQC assumptions remains uncertain.

4.3 Implementation Risks and Side-Channel Vulnerabilities

Theoretical security guarantees are meaningful only when cryptographic systems are implemented securely in practice. PQC algorithms introduce several implementation complexities that increase the risk of side-channel vulnerabilities and practical exploitation.

One major challenge is achieving *constant-time implementation*. Cryptographic software must avoid execution patterns that depend on secret data, since timing variations may leak information about secret keys. Implementing constant-time arithmetic for complex lattice operations and sampling procedures is significantly harder than for traditional modular arithmetic used in RSA or ECC.

Cache-timing attacks exploit memory access patterns to infer secret information, while power analysis attacks monitor fluctuations in power consumption during cryptographic computation. Fault injection attacks deliberately introduce computational errors through

methods such as voltage glitching or electromagnetic interference in order to reveal secret key material.

4.4 Diversity Risk: Over-Reliance on Lattices

A major concern within the current PQC ecosystem is the heavy concentration of standardized algorithms within the lattice-based cryptographic family. Although lattice cryptography currently appears highly promising and resistant to known attacks, this concentration introduces systemic risk.

If a major breakthrough were discovered against lattice problems, a large portion of the standardized PQC ecosystem could become vulnerable simultaneously. Such dependence on a single mathematical family reduces resilience and weakens defense diversity.

The cryptographic community generally benefits from diversity of assumptions. If one mathematical family is compromised, others remain available as fallback options. The current over-reliance on lattice systems therefore represents a long-term strategic concern.

4.5 Theoretical Gaps and Unproven Security

Perhaps the most fundamental limitation of post-quantum cryptography is that its security remains based on computational assumptions rather than unconditional mathematical proof.

Lattice-based systems depend on the presumed hardness of problems such as:

- Learning With Errors (LWE)
- Ring-LWE
- Module-LWE
- Short Integer Solution (SIS)

Although these problems have resisted decades of cryptanalytic effort, there is no formal proof guaranteeing that efficient classical or quantum algorithms do not exist for solving them.

Security estimates are therefore inherently uncertain and depend on current understanding of attack complexity. Improvements in lattice reduction algorithms, BKZ variants, sieving methods, or quantum-enhanced cryptanalysis could weaken existing parameter choices.

Unlike symmetric cryptography, where the impact of Grover's algorithm is relatively well understood, the precise quantum hardness of many structured lattice problems remains unclear. Future advances in quantum algorithm research could potentially reduce security more than currently anticipated.

The collapse of SIKE further illustrates the danger of relying on relatively young mathematical assumptions. Despite years of expert review, hidden structural weaknesses remained undiscovered until a practical attack was demonstrated.

Consequently, confidence in PQC security continues to evolve, and long-term resilience cannot yet be considered fully established.

4.6 Summary Table

Shortcoming	Core Issue	Severity
Size and Performance Overhead	Large keys, ciphertexts, and computational cost	High (Near-Term Impact)
New Attack Surfaces	Structural vulnerabilities and novel algebraic attacks	Medium–High
Implementation Risks	Side-channel leakage and implementation complexity	High
Diversity Risk	Heavy dependence on lattice-based cryptography	Medium (Long-Term Risk)
Migration Complexity	Costly infrastructure retrofitting and hybrid deployment burden	Critical
Theoretical Gaps	Security assumptions remain unproven and evolving	Medium–High

Table 5: Major shortcomings of current post-quantum cryptographic systems

5 Future Prospects

The shortcomings cataloged above define the immediate trajectory of PQC research and deployment.

5.1 Standardisation and Ecosystem Expansion

The NIST roadmap extends beyond the 2024 standards. **FIPS 206 (FN-DSA)** is finalizing compact lattice signatures with 666-byte outputs. In March 2025, **HQC** was selected as a code-based KEM backup, directly addressing lattice concentration risk. Furthermore, NIST’s concurrent "on-ramp" process is soliciting non-lattice signature candidates (SQIsign, MAYO, UOV).

5.2 New Cryptographic Constructions

- **Fully Homomorphic Encryption (FHE):** Schemes like BFV, CKKS, and TFHE are grounded in the same Ring-LWE hardness assumptions as ML-KEM, enabling quantum-resistant computation on encrypted data for cloud privacy and secure ML inference.
- **Post-Quantum ZKPs:** Classical SNARKs rely on pairings vulnerable to Shor’s. Post-quantum replacements constructed from hash commitments (STARKs) or lattice trapdoors are an active high-impact research direction.
- **SQIsign:** Avoiding SIKE’s flaw, SQIsign relies on the Deuring correspondence between isogenies and quaternion ideals, producing signatures of just ~ 177 bytes.

5.3 Technical Optimisations

- **Hardware Acceleration:** Silicon integration of the NTT (e.g., ARM Cortex-M instruction extensions, RISC-V co-processors) will resolve the IoT deployment bottleneck, reducing ML-KEM latency by an order of magnitude.
- **Formal Verification:** Projects like HACL* and EverCrypt use the F* proof assistant to produce machine-checked proofs of functional correctness and constant-time properties, addressing implementation risks at their root.
- **QKD Integration:** Layering computational security (Module-LWE) with information-theoretic security (Quantum Key Distribution) will secure high-sensitivity national networks.

5.4 Mathematical Advances

The long-term security of PQC depends on resolving the quantum complexity of the Shortest Vector Problem. A proof of QMA-hardness would dramatically strengthen confidence in lattices, while a subexponential quantum algorithm for structured ideal lattices would necessitate urgent parameter revisions. The bidirectional relationship between cryptographic needs and pure mathematical inquiry (e.g., algebraic number theory, spectral graph theory) remains the field's most productive dynamic.

6 Conclusion

Post-quantum cryptography has crossed from theoretical research into active global deployment. The NIST standards provide the foundation; however, HQC standardisation, hardware NTT acceleration, formally verified implementations, and continued mathematical scrutiny of lattice and isogeny hardness are the necessary building blocks of the next phase. The harvest-now-decrypt-later clock is already running. The field's future depends on maintaining rigorous, adversarial, interdisciplinary scrutiny to secure the world's digital infrastructure against the quantum horizon.

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